

Ke Zhai

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Machine Learning · Cloud Computing · Natural Language Processing · Audio Processing and Modeling
Large Language Model · Responsible AI and LLM Safety

EDUCATION

Ph.D. in Computer Science

University of Maryland, College Park, MD, 2014

GPA: 3.9/4.0. Supervisor: Jordan Boyd-Graber

Research topic: Large Scale Inference for Probabilistic Bayesian Models

Ph.D. thesis: Models, Inference, and Implementation for Scalable Probabilistic Models of Text

M.Sc in Computer Science

University of Maryland, College Park, MD, 2011

GPA: 3.9/4.0. Supervisor: Jordan Boyd-Graber and Jimmy Lin (co-supervised)

Research topic: Variational Bayesian Inference of Latent Dirichlet Allocation in MapReduce

Master scholarly paper: Using Variational Inference and MapReduce to Scale Topic Modeling

B.Eng. in Computer Engineering

Nanyang Technological University, Singapore, 2009

GPA: 4.65/5.0 with first class honor. Supervisor: Wee Keong Ng

Research topic: Privacy-Preserving Data Mining

Undergraduate thesis: An Embedded Caching Framework for Privacy-Preserving Data Mining

EMPLOYMENT

Staff Research Scientist

Responsible AI Team, Apple, Inc., Cupertino, CA, Oct 2023 - Present

- Founding member of Apple's responsible AI team.
- Infrastructure architect for the team on automatic red-teaming data synthesization, on-device inference/simulation, auto grading and metric reporting platform. The designed system has shown to be highly extensible, customizable, and scalable, which powers all the safety evaluations.
- Tech lead for red teaming efforts for Apple Intelligence features, including web QA, text summarization, text composition and text assistants.
- Tech lead for safety benchmark evaluation on Apple Intelligence models, including harmfulness, responsiveness, biases and fairness.
- Tech lead for developing embedding based semantic override models.

Staff Research Scientist Machine Intelligence Sensing Team, Apple, Inc., Cupertino, CA, Jul 2021 - Oct 2023

- Tech lead for improving handwashing feature for Apple watch. Using audio and motion sensing to improve recall and precisions for handwashing.
- Tech lead for double tap gesture detection for Apple watch, featured in Sep 2023 Apple special event.
- On-device audio detection model for health event for AirPods.
- Location tracking and prediction algorithm based on IMU sensing to optimize energy and battery for Vision Pro.

Founding Member

Dawnlight, Inc., Palo Alto, CA, Jan 2019 - Jul 2021

- Design and develop end-to-end data collection, signal processing, model training and serving architecture (in C++/C) on various IoT edge devices for healthcare related audio event detection.
- Design and develop state-of-the-art audio event detection model (benchmarked on multiple public datasets).
- Design and develop end-to-end data processing, model training and serving architecture for Clinical Decision Support System (CDSS).

- Design and develop end-to-end data processing, model training and feature demo on activity recognition using radar signals and RFID signals.
- Working with a team to design and develop a generic end-to-end machine learning platform with hyper-parameter tuning (using Ray) and container orchestration pipeline support.
- Working with a team to develop and maintain full-stack daily operations, including flask services, bluetooth services, docker containers, Kafka, gRPC, etc.

Senior Research Scientist

Microsoft, Sunnyvale, CA, Aug 2016 - Jan 2019

- Language model personalization using topic models, context matching and style filtering.
- Design and develop end-to-end data processing, model training and adaptation evaluation pipeline.
- Language model adaptation support for Microsoft Teams product for both meeting and broadcasting scenarios. Prototype demo'ed at Microsoft *Build 2018*, *Inspire 2018*, and *Ignite 2018*.

Research Scientist

Yahoo! Labs, Sunnyvale, CA, Feb 2015 - Aug 2016

- Develop query understanding and sequence tagging models for online Ads serving system.
- Design and develop query classification and user intent prediction pipeline for mobile search.
- Research, design and develop on “Chat Bot as a Service” platform.

Graduate Research Assistant Department of Computer Science, University of Maryland, College Park, MD, Sep 2010 - Jun 2014

- Academic Advisor: Jordan Boyd-Graber
- Research Interest: Machine Learning, Non-parametric Bayesian Learning, Cloud Computing
- Design and implement online variational inference for adaptor grammars.
- Design and implement online variational inference for topic models with infinite vocabulary.
- Design and implement variational inference for latent Dirichlet allocation in MapReduce.
- Design and implement variational inference for Indian buffet process in MapReduce.

Research Intern

Yahoo! Labs, New York City, NY, Jun 2014 - Aug 2014

- Large scale unsupervised nonparametric models for user behavior analysis.

Research Intern

Microsoft Research, Redmond, WA, May 2013 - Aug 2013

- Mentor: Jason D. Williams
- Design and implement three models in discovering latent structure in dialogues.
- Achieve comparably well results against many other models on real datasets.

Software Engineering Intern

comScore, Inc., Reston, VA, May 2010 - Aug 2010

- Implement data transfer and formatter block for new deployed Hadoop distributed file system.
- Research and develop cookie deletion and prediction system for comScore.

PUBLICATION (* indicates equal contributor)

Vikramjit Mitra, Anirban Chatterjee, **Ke Zhai**, Helen Weng, Ayuko Hill, Nicole Hay, Christopher Webb, Jamie Cheng and Erdrin Azemi. “Pre-Trained Foundation Model Representations to Uncover Breathing Patterns in Speech”. *23rd International Workshop on Data Mining in Bioinformatics (BioKDD)*, Aug 2024.

Ke Zhai*, and Huan Wang*. “Adaptive Dropout with Rademacher Complexity Regularization”. *International Conference on Learning Representations (ICLR)*, May 2018.

Ke Zhai, Zornitsa Kozareva, Yuening Hu, Qi Li and Weiwei Guo. “Query to Knowledge: Unsupervised Entity Extraction from Shopping Queries using Adaptor Grammars”. *International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, Jul 2016.

Zornitsa Kozareva, Qi Li, **Ke Zhai** and Weiwei Guo. “Recognizing Salient Entities in Shopping Queries”. *Annual Meeting of the Association for Computational Linguistics (ACL)*, Jun 2016.

Ke Zhai, Jordan Boyd-Graber and Shay B. Cohen. “Online Adaptor Grammars with Hybrid Inference”. *Transaction of the Association for Computational Linguistics (TACL)*, Oct 2014.

Ke Zhai, and Jason D. Williams. “Discovering Latent Structure in Task-Oriented Dialogues”. *Annual Meeting of the Association for Computational Linguistics (ACL)*, Jun 2014.

Ke Zhai*, Yuening Hu*, Vladimir Edelman, and Jordan Boyd-Graber. “Polylingual Tree-Based Topic Models for Translation Domain Adaptation”. *Annual Meeting of the Association for Computational Linguistics (ACL)*, Jun 2014.

Ke Zhai, and Jordan Boyd-Graber. “Online Latent Dirichlet Allocation with Infinite Vocabulary”. *International Conference on Machine Learning (ICML)*, Jun 2013.

Ke Zhai*, Yuening Hu*, Jordan Boyd-Graber, and Sinead Williamson. “Modeling Images using Transformed Indian Buffet Processes”. *International Conference on Machine Learning (ICML)*, Jun 2012.

Ke Zhai, Jordan Boyd-Graber, Nima Asadi, and Mohamad Alkhouja. “Mr. LDA: A Flexible Large Scale Topic Modeling Package using Variational Inference in MapReduce”. *ACM International Conference on World Wide Web (WWW)*, Apr 2012.

Ke Zhai, Wee Keong Ng, Andre Ricardo Herianto and Shuguo Han. “Speeding Up Secure Computations via Embedded Caching”. *Proceedings of SIAM International Conference on Data Mining (SDM)*, Apr 2009.

PROFESSIONAL CONTRIBUTION

Online released code-base on github.com/kzhai.

- Activate contributor for MapReduce library Cloud⁹ and Hadoop toolkit Ivory.
- MapReduce latent Dirichlet allocation (Variational Bayesian inference, with extension to informed prior and polylingual LDA).

“I tried both and found that, despite *Mr. LDA*’s cringeworthy name, it’s much faster than *Mahout*’s implementation so decided to go with that one.”

— **Kris Jack**, BSc Hons, Ph.D., Chief Data Scientist, Mendeley

- Latent Dirichlet allocation (Gibbs sampling, variational Bayesian inference and online version).
- Non-parametric Bayesian models (Indian buffet process, hierarchical Dirichlet process and infinite Gaussian mixture model).
- Customizable deep learning toolkits using Theano and PyTorch.